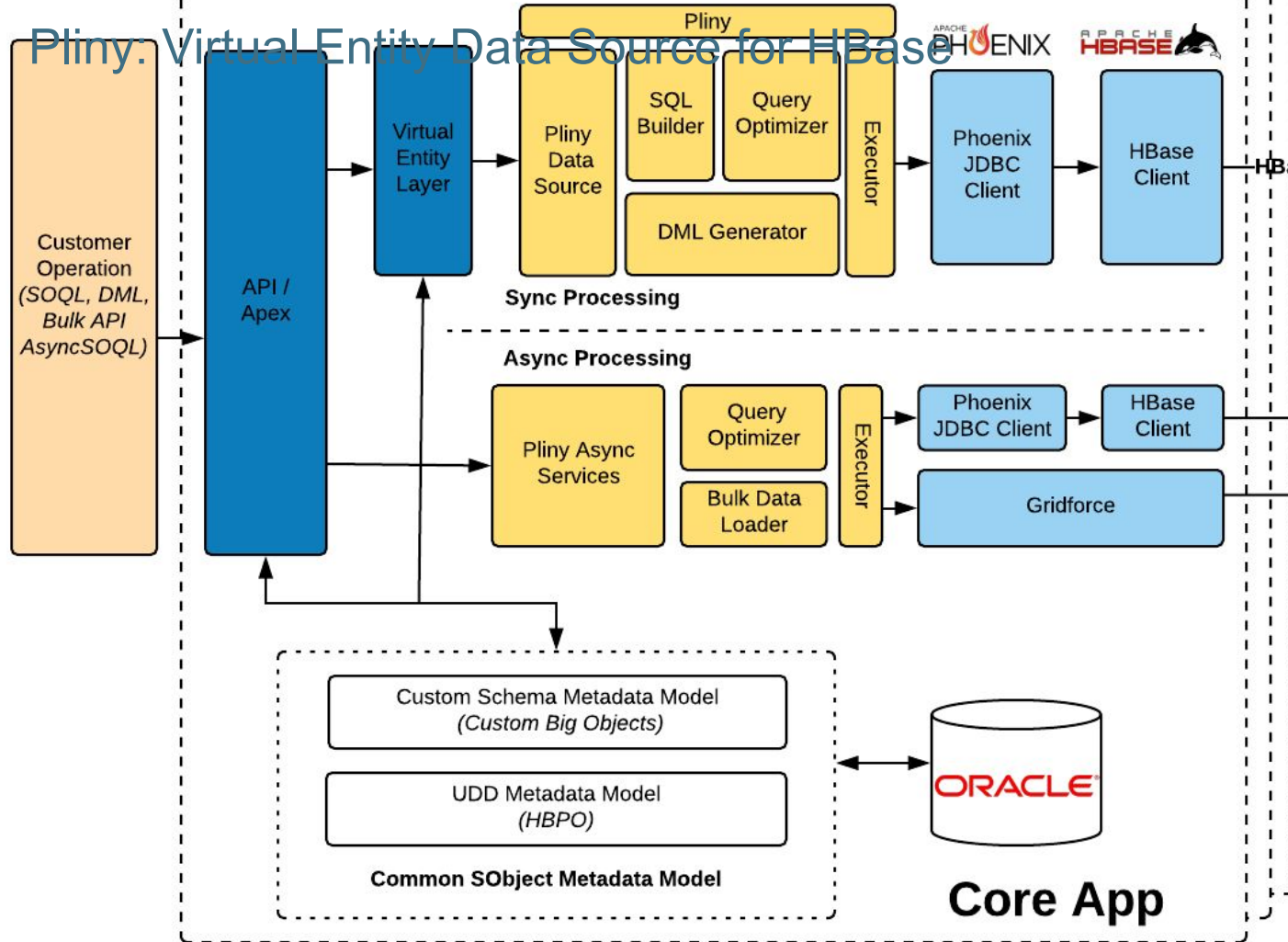
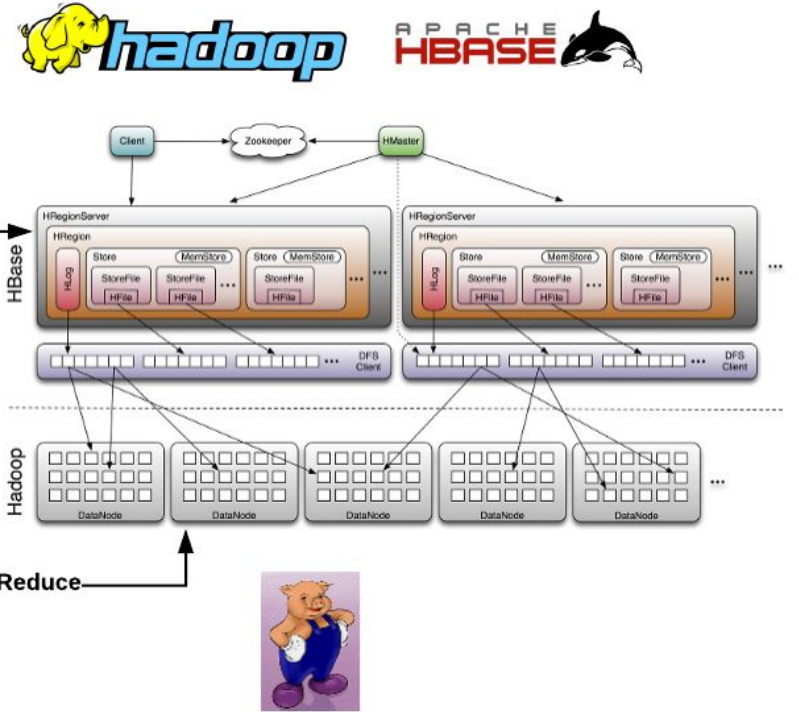


Pod

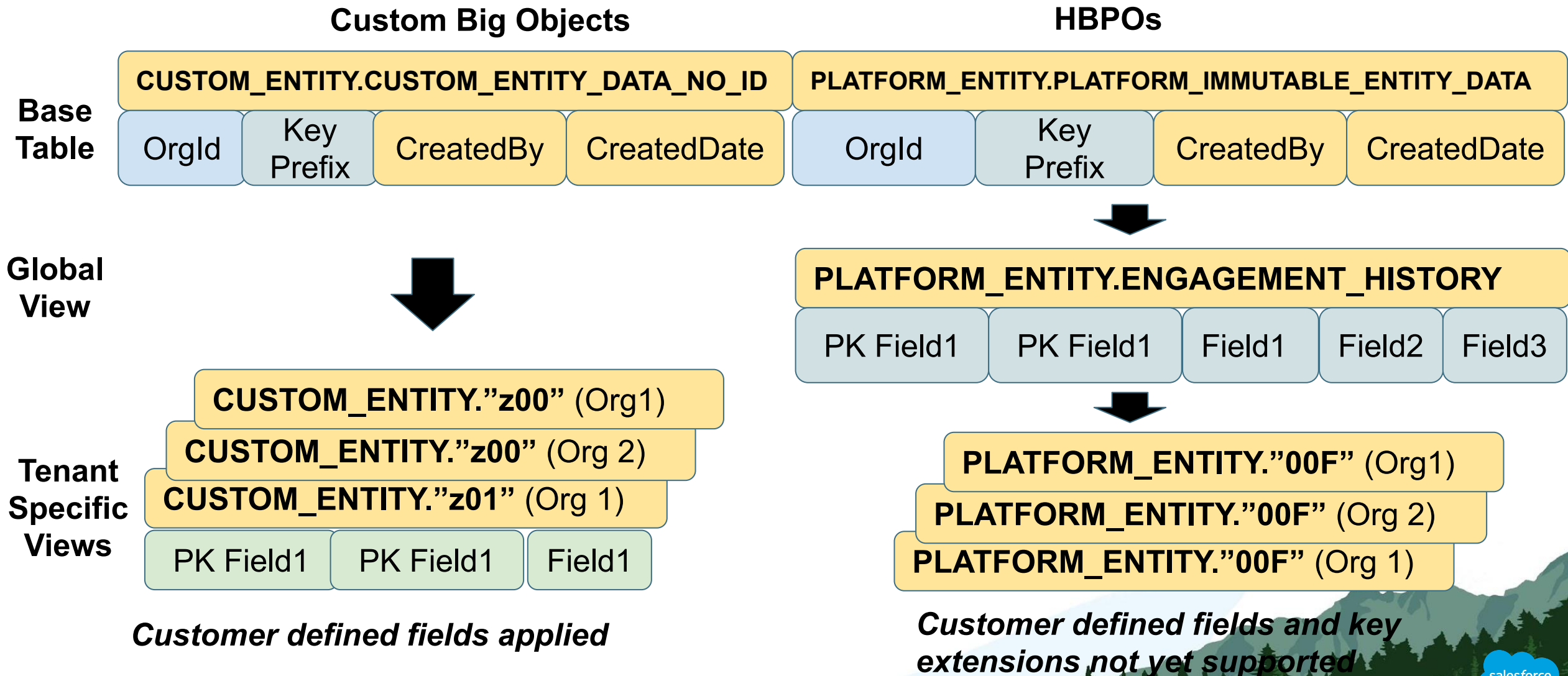


Super Pod



Big Objects: Native Phoenix Views on HBase

Extensible model that makes it easy to work with Big Object data outside app server

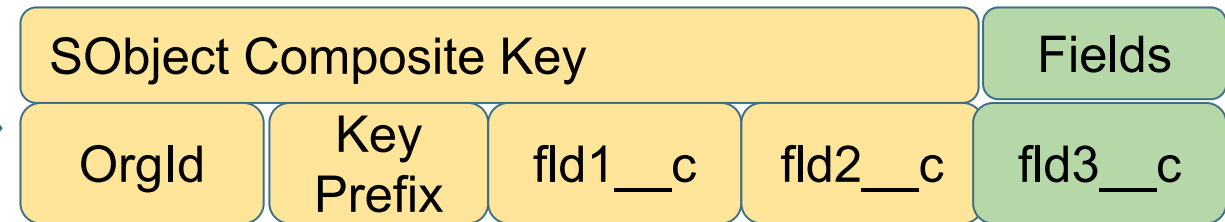


Big Object: Composite Primary Keys

Extending the metadata model to enable performant synch SOQL and idempotency

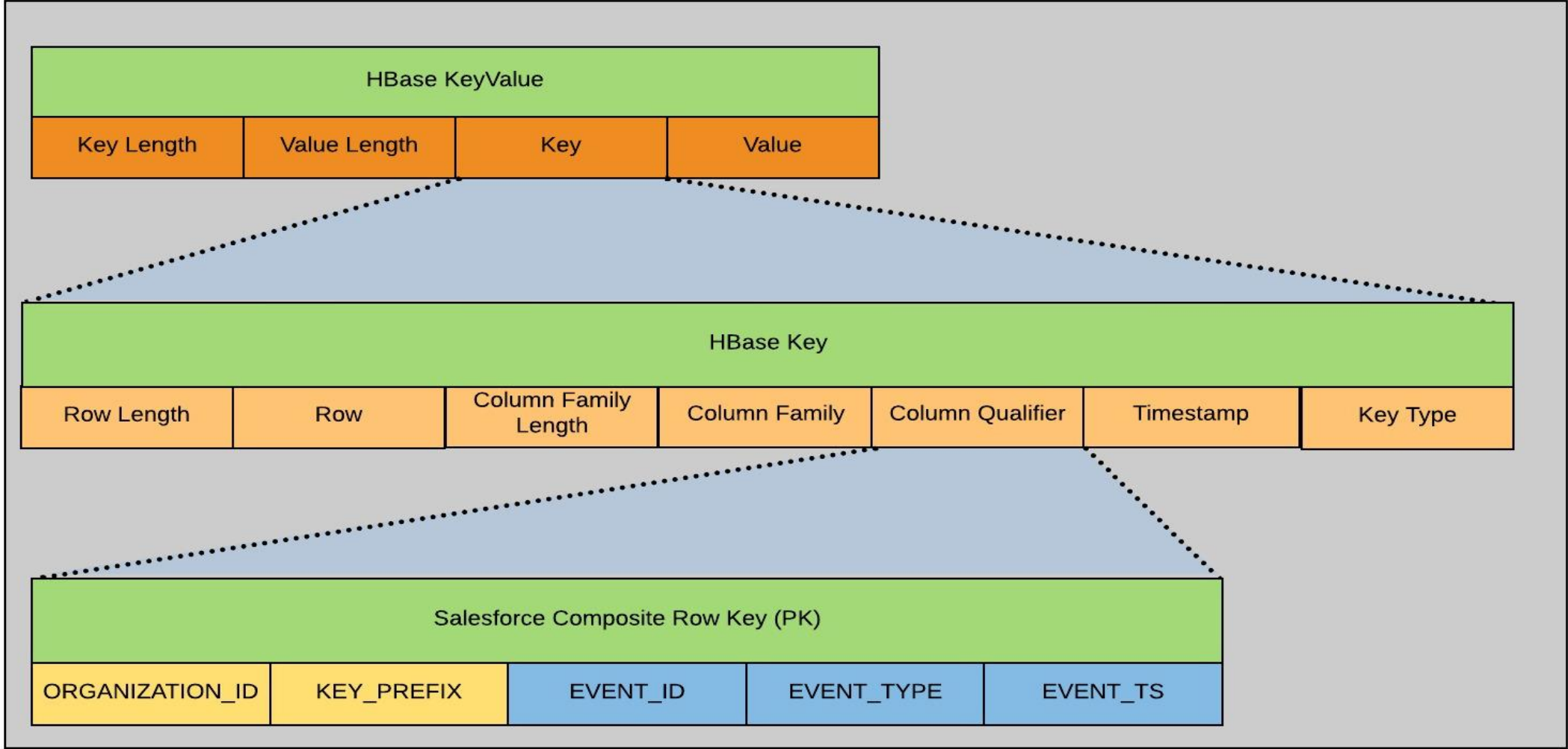
- Access data by key for fast random access - point lookup and bounded scan
- Support for resilient idempotent writes
- New PK index metadata required when defining Big Objects
- Extends existing metadata model

```
...  
<fields> <fullName>idx_fld_1__c</fullName> ... </fields>  
<fields> <fullName>idx_fld_2__c</fullName> ... </fields>  
...  
<indexes>  
  <fields>  
    <name>fld_1__c</name>  
    <sortDirection>DESC</sortDirection>  
  </fields>  
  <fields>  
    <name>fld_2__c</name>  
    <sortDirection>ASC</sortDirection>  
  </fields>  
  <fullName>MyBigObject1PK</fullName>  
  <type>PRIMARY</type>  
</indexes>  
<label>MyBigObject1</label>  
<pluralLabel>MyBigObject1</pluralLabel>  
...
```



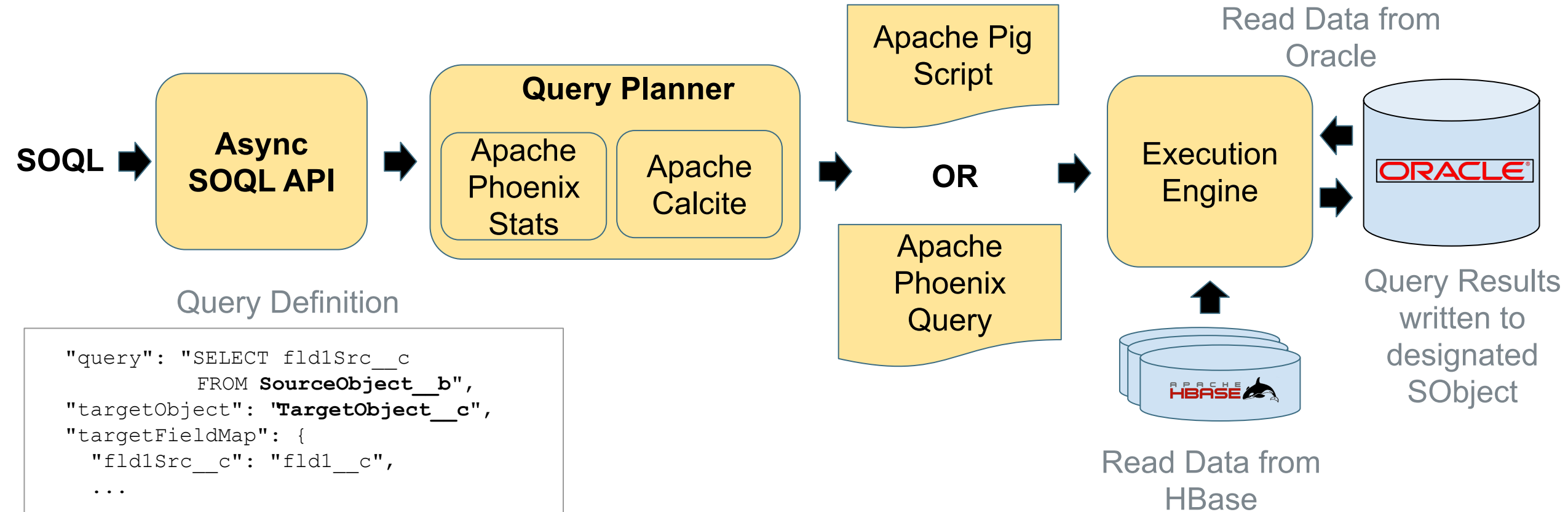
Big Object: Composite Primary Keys

Mapping rowkey metadata to Phoenix and HBase



Async SOQL: Query Pipeline

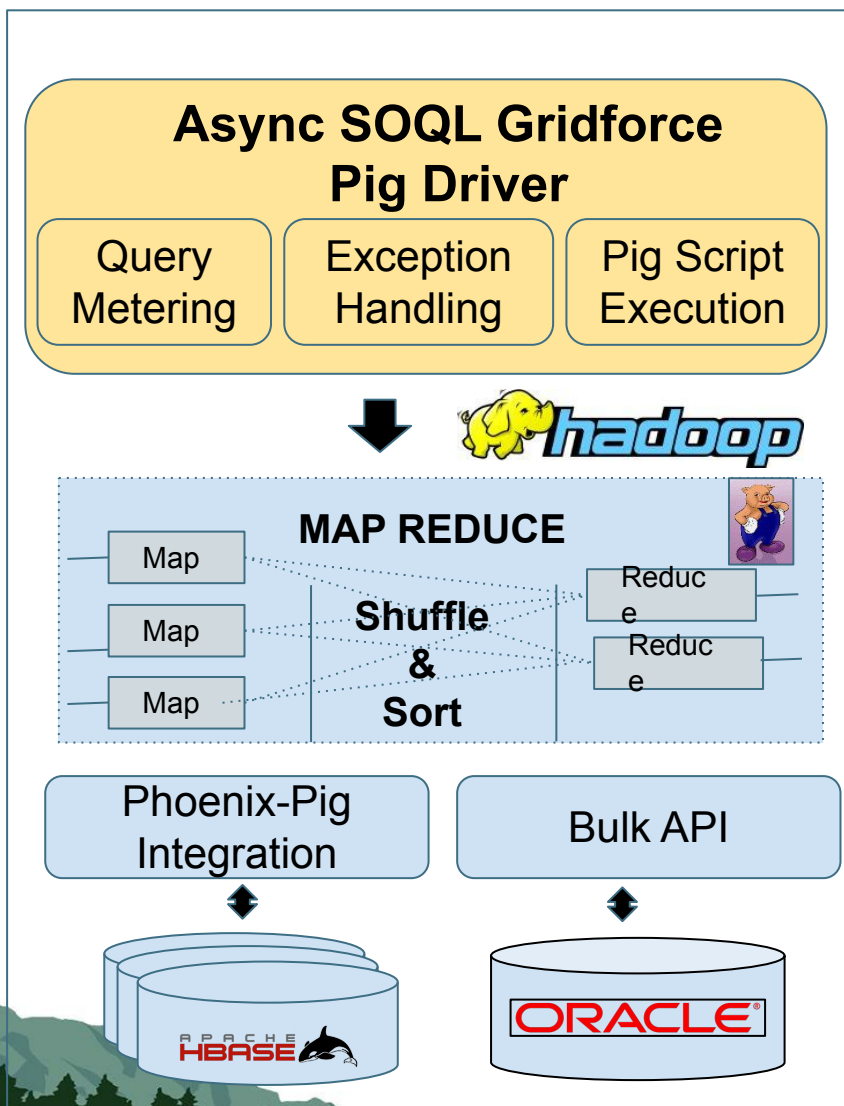
The Async SOQL execution path



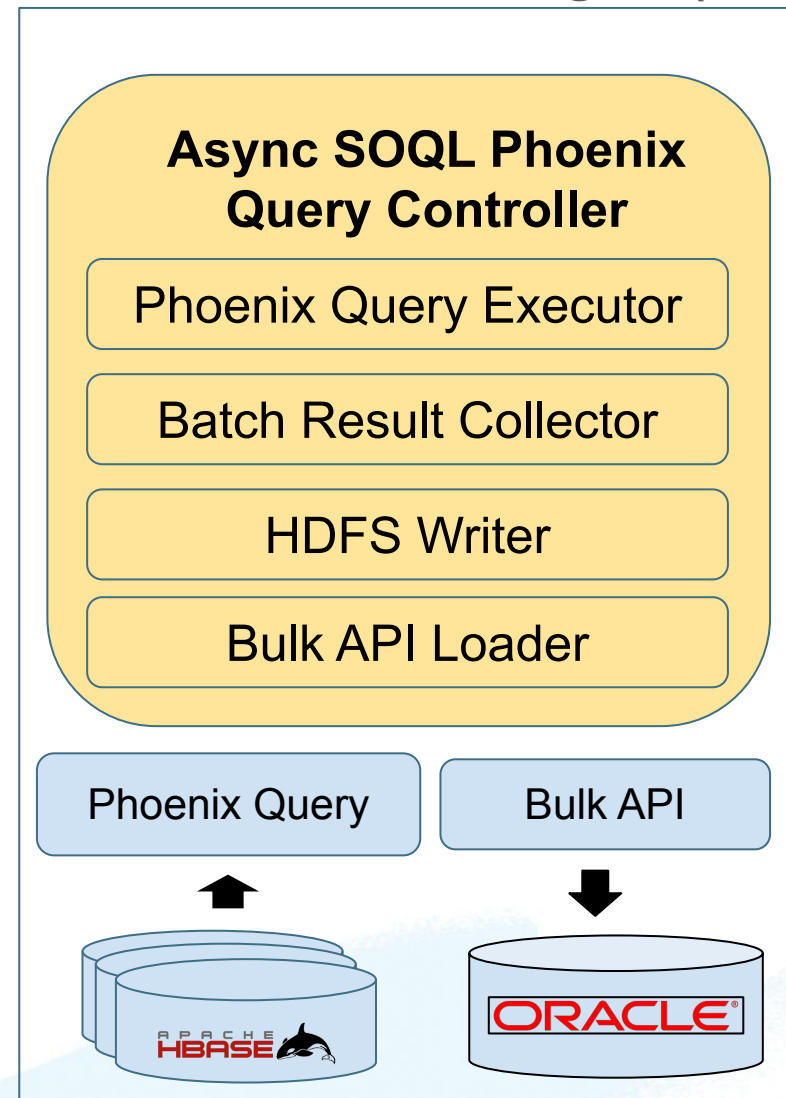
Async SOQL: Engines

Different execution engines invoked based on runtime statistics

Gridforce Apache Pig Engine (GA)



Phoenix Execution Engine (Pilot)



Big Objects: Phoenix Secondary Indexes

Support for alternate query patterns, Limited HBPO support only

In the Big Data world an Index is a copy of your data



Key Characteristics

- HBase doesn't support indexes natively
- Phoenix has support for secondary indexes
- Limited HBPO support for Immutable Global Secondary Indexes
- Clients responsible for index data consistency by ensuring retrying until successful batch
- Expensive storage ($IDX_NUM * Data\ Size$)
- Write Amplification

Big Objects: Phoenix Secondary Indexes

Phoenix Secondary Index Support at SFDC



Type	Platform Usage	Notes
Immutable Global Indexes	HBPOs Only	• Supported at SFDC • Clients responsible for index data consistency, expensive to create, query performance no different than against PK
Mutable Global Indexes	N/A	• Not supported at SFDC (not horizontally scalable) expect for grandfathered in use cases
Local Indexes	Potential future use	• HBase team assessing feasibility of usage at SFDC • Faster write times, write consistency, queries need more threads

Event and Big Object Integration

Work with data in motion and data at rest

